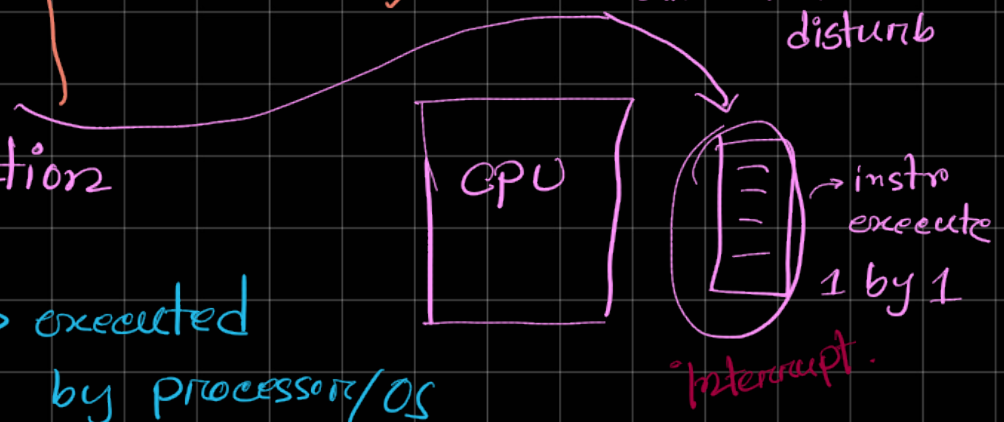
Top level view of computer

CPU
Cache
M. Memory
Hard drive
Peripherals

} seqn. based on speed
Mem hierarchy.

inefficient execution



code segment → executed by processor/OS

I/O segment → " peripherals

program flow without interrupt:
no instruction → so processor sits idle.

soln: processor will pause the flow & leave to where there is instruction.

Dependency: requires value for i/o operation, can't establish segment so jump to another program.

if the i/o operan is done how would the processor be informed?

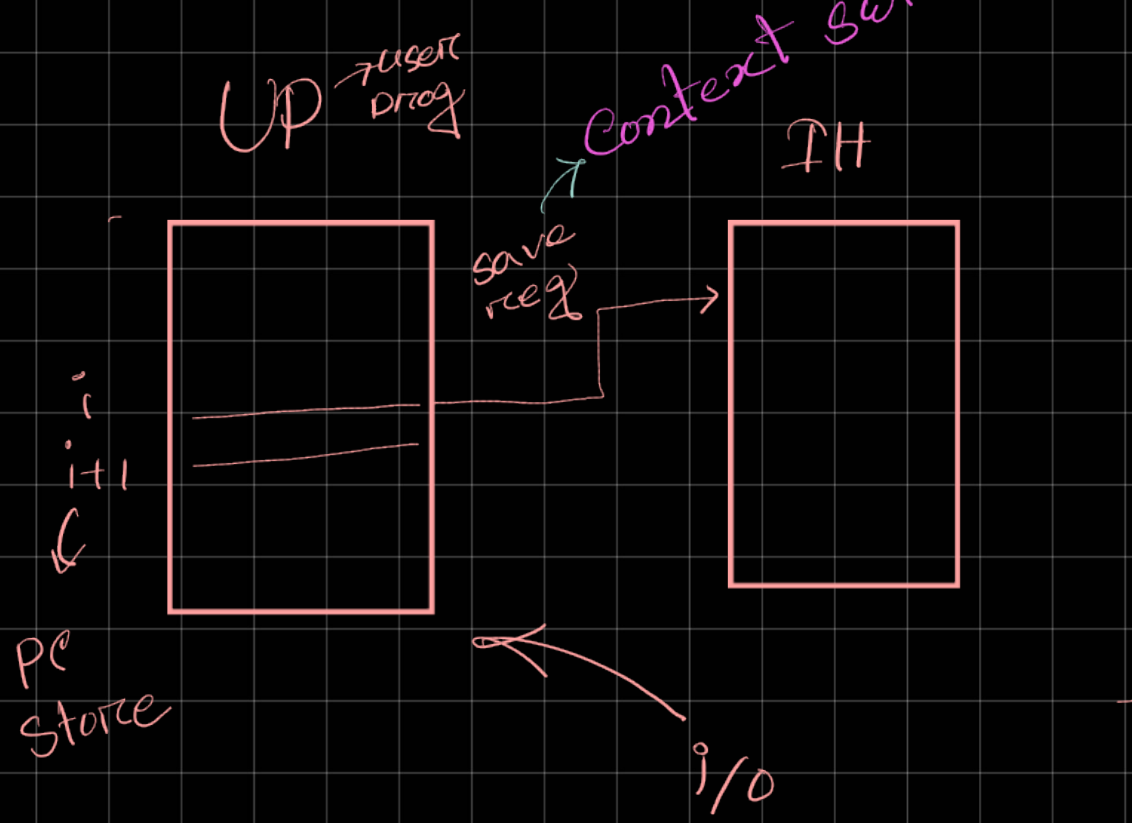
=>

peripheral → executing I/O operation. the ez interruption

IRQ – interrupt req. → CPU respond → Start

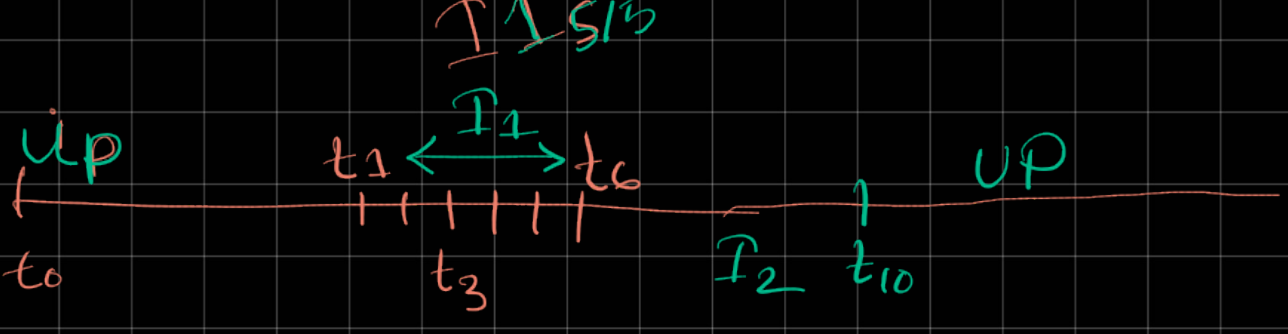
ISR — " Service routine

IH → " Handler



in betw^n instr cycle pie stored in different registers. interruption → image lost

timing diagram:

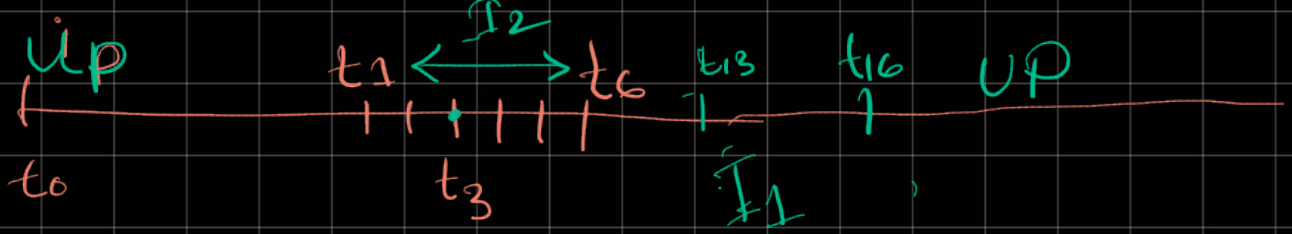


I10/2 → priority 2

↓

10 cycles

I1 S/2



↓

I2 40/5 ⟶ priority 2

↓

10 cycles